

Example Workflow

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Preamble

Peak2Flux2 takes gas chromatography (GC) instrument output to calculate fluxes, specifically in greenhouse gas emissions experiments. An in development package is ready for use on GitHub. We invite you to try out the package and provide feedback to the authors of the package.

We have provided here an example workflow. Here is a data set that has measured methane (CH₄) and nitrous oxide (N₂O)

For questions, please reach out to:

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Instll Peak2Flux2 and load package

```
# Install devtools if not already installed
if (!require("devtools")) install.packages("devtools")
```

```
## Loading required package: devtools
```

```
## Loading required package: usethis
```

```
# Load devtools
library(devtools)
```

```
# Install the GitHub package
install_github("XiaoZhangZhangRice/Peak2Flux2")
```

```
## Using GitHub PAT from the git credential store.
```

```
## Skipping install of 'Peak2Flux2' from a github remote, the SHA1 (be67ee44) has not changed since last
```

```
## Use `force = TRUE` to force installation
```

```
# Note that this package has dependencies (i.e. requires other packages to work). When prompted, please
```

```
# Load the package
```

```
library(Peak2Flux2)
```

```
library(dplyr)
```

```
##
```

```
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
##      filter, lag
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
##      intersect, setdiff, setequal, union
```

Fill up the excel spreadsheets

The package is designed to take a dataframe with a specific input. Please key in your raw data as per specified in the following directory.

There are two files that needs to be populated * Standard calibration file (Std_Input.csv): These are standards and their associated peak areas * Sample file (Input_samples_No_ppm.csv): These are samples and their associated peak areas, metadata used for chamber volume computation and other necessary information.

Note: There is a column called GC_Run. We define this a single continuously run, whereby a standard curve is generated and used to determine sample gas concentrations. The samples will be calibrated by the standards from the same “GC_Run”. This is important and chromatography characteristics change due to considerations such as but not limited to, filter degradation, detector cleanliness (e.g. FID), reagent changes (e.g. nickel 63 in the the ECD). Hence standard curves generated for a single run are unique to that run and cannot be applied to other runs.

Load excel files

The Standard calibration file (Std_Input.csv) and the sample file (Input_samples_No_ppm.csv) on GitHub are located in the directory: XiaoZhangZhangRice/Peak2Flux2/data

```
#Read Standard Calibration File
```

```
std <- read.csv("../data/Std_Input.csv")
```

```
#Read Sample File
```

```
samples <- read.csv("../data/Input_samples_No_ppm.csv")
```

Peak to PPM

This function uses data entries in the standard calibration file to create calibration curves for each individual GC run using peak areas and gas standard ppms. Afterwards, the concentration of samples (ppm) are generated from the peak areas in the samples file and the calibration curves constructed from the standard file.

```
sample_ppm <- peak2ppm(std, samples)
```

```
sample_ppm
```

##	GC_Run	Date	Plot	Treatment	Time_mins
## Batch12_2024.1	Batch12_2024	27-Jun-24	S-M-W-1	SMW	0
## Batch12_2024.2	Batch12_2024	27-Jun-24	S-M-W-1	SMW	21
## Batch12_2024.3	Batch12_2024	27-Jun-24	S-M-W-1	SMW	42
## Batch12_2024.4	Batch12_2024	27-Jun-24	S-M-W-1	SMW	63
## Batch12_2024.5	Batch12_2024	27-Jun-24	C-S-1	CS	0
## Batch12_2024.6	Batch12_2024	27-Jun-24	C-S-1	CS	21
## Batch12_2024.7	Batch12_2024	27-Jun-24	C-S-1	CS	42
## Batch12_2024.8	Batch12_2024	27-Jun-24	C-S-1	CS	63
## Batch12_2024.9	Batch12_2024	27-Jun-24	S-M-1	SM	0
## Batch12_2024.10	Batch12_2024	27-Jun-24	S-M-1	SM	21
## Batch12_2024.11	Batch12_2024	27-Jun-24	S-M-1	SM	42
## Batch12_2024.12	Batch12_2024	27-Jun-24	S-M-1	SM	63
## Batch12_2024.13	Batch12_2024	27-Jun-24	C-S-MD-1	CSMD	0
## Batch12_2024.14	Batch12_2024	27-Jun-24	C-S-MD-1	CSMD	21
## Batch12_2024.15	Batch12_2024	27-Jun-24	C-S-MD-1	CSMD	42
## Batch12_2024.16	Batch12_2024	27-Jun-24	C-S-MD-1	CSMD	63
## Batch12_2024.17	Batch12_2024	27-Jun-24	C-1	C	0
## Batch12_2024.18	Batch12_2024	27-Jun-24	C-1	C	21
## Batch12_2024.19	Batch12_2024	27-Jun-24	C-1	C	42
## Batch12_2024.20	Batch12_2024	27-Jun-24	C-1	C	63
## Batch12_2024.21	Batch12_2024	27-Jun-24	C-S-MD-2	CSMD	0
## Batch12_2024.22	Batch12_2024	27-Jun-24	C-S-MD-2	CSMD	21
## Batch12_2024.23	Batch12_2024	27-Jun-24	C-S-MD-2	CSMD	42
## Batch12_2024.24	Batch12_2024	27-Jun-24	C-S-MD-2	CSMD	63
## Batch12_2024.25	Batch12_2024	27-Jun-24	C-2	C	0
## Batch12_2024.26	Batch12_2024	27-Jun-24	C-2	C	21
## Batch12_2024.27	Batch12_2024	27-Jun-24	C-2	C	42
## Batch12_2024.28	Batch12_2024	27-Jun-24	C-2	C	63
## Batch12_2024.29	Batch12_2024	27-Jun-24	C-S-2	CS	0
## Batch12_2024.30	Batch12_2024	27-Jun-24	C-S-2	CS	21
## Batch12_2024.31	Batch12_2024	27-Jun-24	C-S-2	CS	42
## Batch12_2024.32	Batch12_2024	27-Jun-24	C-S-2	CS	63
## Batch12_2024.33	Batch12_2024	27-Jun-24	S-M-2	SM	0
## Batch12_2024.34	Batch12_2024	27-Jun-24	S-M-2	SM	21
## Batch12_2024.35	Batch12_2024	27-Jun-24	S-M-2	SM	42
## Batch12_2024.36	Batch12_2024	27-Jun-24	S-M-2	SM	63
## Batch12_2024.37	Batch12_2024	27-Jun-24	S-M-W-2	SMW	0
## Batch12_2024.38	Batch12_2024	27-Jun-24	S-M-W-2	SMW	21
## Batch12_2024.39	Batch12_2024	27-Jun-24	S-M-W-2	SMW	42
## Batch12_2024.40	Batch12_2024	27-Jun-24	S-M-W-2	SMW	63
## Batch12_2024.41	Batch12_2024	27-Jun-24	C-3	C	0
## Batch12_2024.42	Batch12_2024	27-Jun-24	C-3	C	21
## Batch12_2024.43	Batch12_2024	27-Jun-24	C-3	C	42
## Batch12_2024.44	Batch12_2024	27-Jun-24	C-3	C	63
## Batch12_2024.45	Batch12_2024	27-Jun-24	C-S-3	CS	0
## Batch12_2024.46	Batch12_2024	27-Jun-24	C-S-3	CS	21
## Batch12_2024.47	Batch12_2024	27-Jun-24	C-S-3	CS	42
## Batch12_2024.48	Batch12_2024	27-Jun-24	C-S-3	CS	63
## Batch12_2024.49	Batch12_2024	27-Jun-24	S-M-3	SM	0
## Batch12_2024.50	Batch12_2024	27-Jun-24	S-M-3	SM	21
## Batch12_2024.51	Batch12_2024	27-Jun-24	S-M-3	SM	42
## Batch12_2024.52	Batch12_2024	27-Jun-24	S-M-3	SM	63
## Batch12_2024.53	Batch12_2024	27-Jun-24	S-M-W-3	SMW	0

## Batch12_2024.54	Batch12_2024	27-Jun-24	S-M-W-3	SMW	21
## Batch12_2024.55	Batch12_2024	27-Jun-24	S-M-W-3	SMW	42
## Batch12_2024.56	Batch12_2024	27-Jun-24	S-M-W-3	SMW	63
## Batch12_2024.57	Batch12_2024	27-Jun-24	C-S-MD-3	CSMD	0
## Batch12_2024.58	Batch12_2024	27-Jun-24	C-S-MD-3	CSMD	21
## Batch12_2024.59	Batch12_2024	27-Jun-24	C-S-MD-3	CSMD	42
## Batch12_2024.60	Batch12_2024	27-Jun-24	C-S-MD-3	CSMD	63
## Batch12_2024.61	Batch12_2024	27-Jun-24	C-S-MD-4	CSMD	0
## Batch12_2024.62	Batch12_2024	27-Jun-24	C-S-MD-4	CSMD	21
## Batch12_2024.63	Batch12_2024	27-Jun-24	C-S-MD-4	CSMD	42
## Batch12_2024.64	Batch12_2024	27-Jun-24	C-S-MD-4	CSMD	63
## Batch12_2024.65	Batch12_2024	27-Jun-24	C-4	C	0
## Batch12_2024.66	Batch12_2024	27-Jun-24	C-4	C	21
## Batch12_2024.67	Batch12_2024	27-Jun-24	C-4	C	42
## Batch12_2024.68	Batch12_2024	27-Jun-24	C-4	C	63
## Batch12_2024.69	Batch12_2024	27-Jun-24	S-M-W-4	SMW	0
## Batch12_2024.70	Batch12_2024	27-Jun-24	S-M-W-4	SMW	21
## Batch12_2024.71	Batch12_2024	27-Jun-24	S-M-W-4	SMW	42
## Batch12_2024.72	Batch12_2024	27-Jun-24	S-M-W-4	SMW	63
## Batch12_2024.73	Batch12_2024	27-Jun-24	S-M-4	SM	0
## Batch12_2024.74	Batch12_2024	27-Jun-24	S-M-4	SM	21
## Batch12_2024.75	Batch12_2024	27-Jun-24	S-M-4	SM	42
## Batch12_2024.76	Batch12_2024	27-Jun-24	S-M-4	SM	63
## Batch12_2024.77	Batch12_2024	27-Jun-24	C-S-4	CS	0
## Batch12_2024.78	Batch12_2024	27-Jun-24	C-S-4	CS	21
## Batch12_2024.79	Batch12_2024	27-Jun-24	C-S-4	CS	42
## Batch12_2024.80	Batch12_2024	27-Jun-24	C-S-4	CS	63
## Batch14_2024.81	Batch14_2024	01-Aug-24	S-M-W-1	SMW	0
## Batch14_2024.82	Batch14_2024	01-Aug-24	S-M-W-1	SMW	21
## Batch14_2024.83	Batch14_2024	01-Aug-24	S-M-W-1	SMW	42
## Batch14_2024.84	Batch14_2024	01-Aug-24	S-M-W-1	SMW	63
## Batch14_2024.85	Batch14_2024	01-Aug-24	C-S-1	CS	0
## Batch14_2024.86	Batch14_2024	01-Aug-24	C-S-1	CS	21
## Batch14_2024.87	Batch14_2024	01-Aug-24	C-S-1	CS	42
## Batch14_2024.88	Batch14_2024	01-Aug-24	C-S-1	CS	63
## Batch14_2024.89	Batch14_2024	01-Aug-24	S-M-1	SM	0
## Batch14_2024.90	Batch14_2024	01-Aug-24	S-M-1	SM	21
## Batch14_2024.91	Batch14_2024	01-Aug-24	S-M-1	SM	42
## Batch14_2024.92	Batch14_2024	01-Aug-24	S-M-1	SM	63
## Batch14_2024.93	Batch14_2024	01-Aug-24	C-S-MD-1	CSMD	0
## Batch14_2024.94	Batch14_2024	01-Aug-24	C-S-MD-1	CSMD	21
## Batch14_2024.95	Batch14_2024	01-Aug-24	C-S-MD-1	CSMD	42
## Batch14_2024.96	Batch14_2024	01-Aug-24	C-S-MD-1	CSMD	63
## Batch14_2024.97	Batch14_2024	01-Aug-24	C-1	C	0
## Batch14_2024.98	Batch14_2024	01-Aug-24	C-1	C	21
## Batch14_2024.99	Batch14_2024	01-Aug-24	C-1	C	42
## Batch14_2024.100	Batch14_2024	01-Aug-24	C-1	C	63
## Batch14_2024.101	Batch14_2024	01-Aug-24	C-S-MD-2	CSMD	0
## Batch14_2024.102	Batch14_2024	01-Aug-24	C-S-MD-2	CSMD	21
## Batch14_2024.103	Batch14_2024	01-Aug-24	C-S-MD-2	CSMD	42
## Batch14_2024.104	Batch14_2024	01-Aug-24	C-S-MD-2	CSMD	63
## Batch14_2024.105	Batch14_2024	01-Aug-24	C-2	C	0
## Batch14_2024.106	Batch14_2024	01-Aug-24	C-2	C	21
## Batch14_2024.107	Batch14_2024	01-Aug-24	C-2	C	42

##	Batch14_2024.108	Batch14_2024	01-Aug-24	C-2	C	63
##	Batch14_2024.109	Batch14_2024	01-Aug-24	C-S-2	CS	0
##	Batch14_2024.110	Batch14_2024	01-Aug-24	C-S-2	CS	21
##	Batch14_2024.111	Batch14_2024	01-Aug-24	C-S-2	CS	42
##	Batch14_2024.112	Batch14_2024	01-Aug-24	C-S-2	CS	63
##	Batch14_2024.113	Batch14_2024	01-Aug-24	S-M-2	SM	0
##	Batch14_2024.114	Batch14_2024	01-Aug-24	S-M-2	SM	21
##	Batch14_2024.115	Batch14_2024	01-Aug-24	S-M-2	SM	42
##	Batch14_2024.116	Batch14_2024	01-Aug-24	S-M-2	SM	63
##	Batch14_2024.117	Batch14_2024	01-Aug-24	S-M-W-2	SMW	0
##	Batch14_2024.118	Batch14_2024	01-Aug-24	S-M-W-2	SMW	21
##	Batch14_2024.119	Batch14_2024	01-Aug-24	S-M-W-2	SMW	42
##	Batch14_2024.120	Batch14_2024	01-Aug-24	S-M-W-2	SMW	63
##	Batch14_2024.121	Batch14_2024	01-Aug-24	C-3	C	0
##	Batch14_2024.122	Batch14_2024	01-Aug-24	C-3	C	21
##	Batch14_2024.123	Batch14_2024	01-Aug-24	C-3	C	42
##	Batch14_2024.124	Batch14_2024	01-Aug-24	C-3	C	63
##	Batch14_2024.125	Batch14_2024	01-Aug-24	C-S-3	CS	0
##	Batch14_2024.126	Batch14_2024	01-Aug-24	C-S-3	CS	21
##	Batch14_2024.127	Batch14_2024	01-Aug-24	C-S-3	CS	42
##	Batch14_2024.128	Batch14_2024	01-Aug-24	C-S-3	CS	63
##	Batch14_2024.129	Batch14_2024	01-Aug-24	S-M-3	SM	0
##	Batch14_2024.130	Batch14_2024	01-Aug-24	S-M-3	SM	21
##	Batch14_2024.131	Batch14_2024	01-Aug-24	S-M-3	SM	42
##	Batch14_2024.132	Batch14_2024	01-Aug-24	S-M-3	SM	63
##	Batch14_2024.133	Batch14_2024	01-Aug-24	S-M-W-3	SMW	0
##	Batch14_2024.134	Batch14_2024	01-Aug-24	S-M-W-3	SMW	21
##	Batch14_2024.135	Batch14_2024	01-Aug-24	S-M-W-3	SMW	42
##	Batch14_2024.136	Batch14_2024	01-Aug-24	S-M-W-3	SMW	63
##	Batch14_2024.137	Batch14_2024	01-Aug-24	C-S-MD-3	CSMD	0
##	Batch14_2024.138	Batch14_2024	01-Aug-24	C-S-MD-3	CSMD	21
##	Batch14_2024.139	Batch14_2024	01-Aug-24	C-S-MD-3	CSMD	42
##	Batch14_2024.140	Batch14_2024	01-Aug-24	C-S-MD-3	CSMD	63
##	Batch14_2024.141	Batch14_2024	01-Aug-24	C-S-MD-4	CSMD	0
##	Batch14_2024.142	Batch14_2024	01-Aug-24	C-S-MD-4	CSMD	21
##	Batch14_2024.143	Batch14_2024	01-Aug-24	C-S-MD-4	CSMD	42
##	Batch14_2024.144	Batch14_2024	01-Aug-24	C-S-MD-4	CSMD	63
##	Batch14_2024.145	Batch14_2024	01-Aug-24	C-4	C	0
##	Batch14_2024.146	Batch14_2024	01-Aug-24	C-4	C	21
##	Batch14_2024.147	Batch14_2024	01-Aug-24	C-4	C	42
##	Batch14_2024.148	Batch14_2024	01-Aug-24	C-4	C	63
##	Batch14_2024.149	Batch14_2024	01-Aug-24	S-M-W-4	SMW	0
##	Batch14_2024.150	Batch14_2024	01-Aug-24	S-M-W-4	SMW	21
##	Batch14_2024.151	Batch14_2024	01-Aug-24	S-M-W-4	SMW	42
##	Batch14_2024.152	Batch14_2024	01-Aug-24	S-M-W-4	SMW	63
##	Batch14_2024.153	Batch14_2024	01-Aug-24	S-M-4	SM	0
##	Batch14_2024.154	Batch14_2024	01-Aug-24	S-M-4	SM	21
##	Batch14_2024.155	Batch14_2024	01-Aug-24	S-M-4	SM	42
##	Batch14_2024.156	Batch14_2024	01-Aug-24	S-M-4	SM	63
##	Batch14_2024.157	Batch14_2024	01-Aug-24	C-S-4	CS	0
##	Batch14_2024.158	Batch14_2024	01-Aug-24	C-S-4	CS	21
##	Batch14_2024.159	Batch14_2024	01-Aug-24	C-S-4	CS	42
##	Batch14_2024.160	Batch14_2024	01-Aug-24	C-S-4	CS	63
##		Chamber_Temp_C	Surface_Area_m2	Height_m	Volume_m3	

## Batch12_2024.1	32.3	0.06834928	0.32475	NA
## Batch12_2024.2	31.7	0.06834928	0.32475	NA
## Batch12_2024.3	33.0	0.06834928	0.32475	NA
## Batch12_2024.4	32.5	0.06834928	0.32475	NA
## Batch12_2024.5	33.7	0.06834928	0.32200	NA
## Batch12_2024.6	34.7	0.06834928	0.32200	NA
## Batch12_2024.7	34.8	0.06834928	0.32200	NA
## Batch12_2024.8	34.5	0.06834928	0.32200	NA
## Batch12_2024.9	33.7	0.06834928	0.30600	NA
## Batch12_2024.10	33.4	0.06834928	0.30600	NA
## Batch12_2024.11	33.2	0.06834928	0.30600	NA
## Batch12_2024.12	33.0	0.06834928	0.30600	NA
## Batch12_2024.13	31.6	0.06834928	0.31100	NA
## Batch12_2024.14	32.1	0.06834928	0.31100	NA
## Batch12_2024.15	31.8	0.06834928	0.31100	NA
## Batch12_2024.16	31.6	0.06834928	0.31100	NA
## Batch12_2024.17	33.0	0.06834928	0.29600	NA
## Batch12_2024.18	33.0	0.06834928	0.29600	NA
## Batch12_2024.19	33.3	0.06834928	0.29600	NA
## Batch12_2024.20	33.0	0.06834928	0.29600	NA
## Batch12_2024.21	30.0	0.06834928	0.31000	NA
## Batch12_2024.22	29.9	0.06834928	0.31000	NA
## Batch12_2024.23	29.7	0.06834928	0.31000	NA
## Batch12_2024.24	29.3	0.06834928	0.31000	NA
## Batch12_2024.25	32.1	0.06834928	0.29800	NA
## Batch12_2024.26	30.9	0.06834928	0.29800	NA
## Batch12_2024.27	31.8	0.06834928	0.29800	NA
## Batch12_2024.28	31.9	0.06834928	0.29800	NA
## Batch12_2024.29	32.1	0.06834928	0.30500	NA
## Batch12_2024.30	30.4	0.06834928	0.30500	NA
## Batch12_2024.31	32.5	0.06834928	0.30500	NA
## Batch12_2024.32	32.7	0.06834928	0.30500	NA
## Batch12_2024.33	31.2	0.06834928	0.30650	NA
## Batch12_2024.34	29.2	0.06834928	0.30650	NA
## Batch12_2024.35	31.0	0.06834928	0.30650	NA
## Batch12_2024.36	29.8	0.06834928	0.30650	NA
## Batch12_2024.37	30.4	0.06834928	0.27400	NA
## Batch12_2024.38	29.8	0.06834928	0.27400	NA
## Batch12_2024.39	31.8	0.06834928	0.27400	NA
## Batch12_2024.40	32.1	0.06834928	0.27400	NA
## Batch12_2024.41	29.1	0.06834928	0.30650	NA
## Batch12_2024.42	31.3	0.06834928	0.30650	NA
## Batch12_2024.43	32.2	0.06834928	0.30650	NA
## Batch12_2024.44	31.9	0.06834928	0.30650	NA
## Batch12_2024.45	38.6	0.06834928	0.30550	NA
## Batch12_2024.46	37.1	0.06834928	0.30550	NA
## Batch12_2024.47	40.0	0.06834928	0.30550	NA
## Batch12_2024.48	33.7	0.06834928	0.30550	NA
## Batch12_2024.49	31.1	0.06834928	0.30300	NA
## Batch12_2024.50	32.4	0.06834928	0.30300	NA
## Batch12_2024.51	32.6	0.06834928	0.30300	NA
## Batch12_2024.52	32.4	0.06834928	0.30300	NA
## Batch12_2024.53	30.4	0.06834928	0.30100	NA
## Batch12_2024.54	32.6	0.06834928	0.30100	NA

## Batch12_2024.55	32.5	0.06834928	0.30100	NA
## Batch12_2024.56	31.4	0.06834928	0.30100	NA
## Batch12_2024.57	31.9	0.06834928	0.30750	NA
## Batch12_2024.58	32.7	0.06834928	0.30750	NA
## Batch12_2024.59	32.9	0.06834928	0.30750	NA
## Batch12_2024.60	32.9	0.06834928	0.30750	NA
## Batch12_2024.61	29.4	0.06834928	0.31050	NA
## Batch12_2024.62	29.7	0.06834928	0.31050	NA
## Batch12_2024.63	29.7	0.06834928	0.31050	NA
## Batch12_2024.64	29.9	0.06834928	0.31050	NA
## Batch12_2024.65	27.6	0.06834928	0.30225	NA
## Batch12_2024.66	30.3	0.06834928	0.30225	NA
## Batch12_2024.67	30.3	0.06834928	0.30225	NA
## Batch12_2024.68	29.4	0.06834928	0.30225	NA
## Batch12_2024.69	28.4	0.06834928	0.30050	NA
## Batch12_2024.70	30.5	0.06834928	0.30050	NA
## Batch12_2024.71	34.2	0.06834928	0.30050	NA
## Batch12_2024.72	31.5	0.06834928	0.30050	NA
## Batch12_2024.73	27.8	0.06834928	0.30150	NA
## Batch12_2024.74	31.5	0.06834928	0.30150	NA
## Batch12_2024.75	31.6	0.06834928	0.30150	NA
## Batch12_2024.76	30.1	0.06834928	0.30150	NA
## Batch12_2024.77	29.6	0.06834928	0.30150	NA
## Batch12_2024.78	30.6	0.06834928	0.30150	NA
## Batch12_2024.79	32.0	0.06834928	0.30150	NA
## Batch12_2024.80	30.6	0.06834928	0.30150	NA
## Batch14_2024.81	35.2	0.06834928	0.81070	NA
## Batch14_2024.82	36.3	0.06834928	0.81070	NA
## Batch14_2024.83	36.9	0.06834928	0.81070	NA
## Batch14_2024.84	37.8	0.06834928	0.81070	NA
## Batch14_2024.85	33.9	0.06834928	0.80870	NA
## Batch14_2024.86	36.3	0.06834928	0.80870	NA
## Batch14_2024.87	37.9	0.06834928	0.80870	NA
## Batch14_2024.88	38.2	0.06834928	0.80870	NA
## Batch14_2024.89	36.0	0.06834928	0.79570	NA
## Batch14_2024.90	34.4	0.06834928	0.79570	NA
## Batch14_2024.91	35.9	0.06834928	0.79570	NA
## Batch14_2024.92	35.0	0.06834928	0.79570	NA
## Batch14_2024.93	36.0	0.06834928	0.79645	NA
## Batch14_2024.94	36.6	0.06834928	0.79645	NA
## Batch14_2024.95	39.1	0.06834928	0.79645	NA
## Batch14_2024.96	39.1	0.06834928	0.79645	NA
## Batch14_2024.97	39.9	0.06834928	0.78145	NA
## Batch14_2024.98	40.9	0.06834928	0.78145	NA
## Batch14_2024.99	42.9	0.06834928	0.78145	NA
## Batch14_2024.100	42.7	0.06834928	0.78145	NA
## Batch14_2024.101	34.6	0.06834928	0.79495	NA
## Batch14_2024.102	36.1	0.06834928	0.79495	NA
## Batch14_2024.103	36.3	0.06834928	0.79495	NA
## Batch14_2024.104	37.8	0.06834928	0.79495	NA
## Batch14_2024.105	36.4	0.06834928	0.78420	NA
## Batch14_2024.106	38.9	0.06834928	0.78420	NA
## Batch14_2024.107	39.7	0.06834928	0.78420	NA
## Batch14_2024.108	41.0	0.06834928	0.78420	NA

## Batch14_2024.109	35.6	0.06834928	0.79020	NA
## Batch14_2024.110	37.7	0.06834928	0.79020	NA
## Batch14_2024.111	37.5	0.06834928	0.79020	NA
## Batch14_2024.112	38.1	0.06834928	0.79020	NA
## Batch14_2024.113	36.8	0.06834928	0.79695	NA
## Batch14_2024.114	37.4	0.06834928	0.79695	NA
## Batch14_2024.115	38.3	0.06834928	0.79695	NA
## Batch14_2024.116	38.3	0.06834928	0.79695	NA
## Batch14_2024.117	40.8	0.06834928	0.75945	NA
## Batch14_2024.118	42.0	0.06834928	0.75945	NA
## Batch14_2024.119	41.7	0.06834928	0.75945	NA
## Batch14_2024.120	43.1	0.06834928	0.75945	NA
## Batch14_2024.121	37.5	0.06834928	0.79595	NA
## Batch14_2024.122	40.7	0.06834928	0.79595	NA
## Batch14_2024.123	40.8	0.06834928	0.79595	NA
## Batch14_2024.124	42.4	0.06834928	0.79595	NA
## Batch14_2024.125	37.1	0.06834928	0.79570	NA
## Batch14_2024.126	37.3	0.06834928	0.79570	NA
## Batch14_2024.127	37.9	0.06834928	0.79570	NA
## Batch14_2024.128	39.5	0.06834928	0.79570	NA
## Batch14_2024.129	33.6	0.06834928	0.79270	NA
## Batch14_2024.130	37.8	0.06834928	0.79270	NA
## Batch14_2024.131	38.1	0.06834928	0.79270	NA
## Batch14_2024.132	39.4	0.06834928	0.79270	NA
## Batch14_2024.133	39.1	0.06834928	0.78895	NA
## Batch14_2024.134	39.9	0.06834928	0.78895	NA
## Batch14_2024.135	41.2	0.06834928	0.78895	NA
## Batch14_2024.136	41.5	0.06834928	0.78895	NA
## Batch14_2024.137	36.1	0.06834928	0.79770	NA
## Batch14_2024.138	38.1	0.06834928	0.79770	NA
## Batch14_2024.139	40.1	0.06834928	0.79770	NA
## Batch14_2024.140	41.0	0.06834928	0.79770	NA
## Batch14_2024.141	34.9	0.06834928	0.79970	NA
## Batch14_2024.142	35.6	0.06834928	0.79970	NA
## Batch14_2024.143	38.2	0.06834928	0.79970	NA
## Batch14_2024.144	41.2	0.06834928	0.79970	NA
## Batch14_2024.145	38.9	0.06834928	0.78970	NA
## Batch14_2024.146	37.8	0.06834928	0.78970	NA
## Batch14_2024.147	39.0	0.06834928	0.78970	NA
## Batch14_2024.148	37.6	0.06834928	0.78970	NA
## Batch14_2024.149	36.2	0.06834928	0.78995	NA
## Batch14_2024.150	36.6	0.06834928	0.78995	NA
## Batch14_2024.151	36.8	0.06834928	0.78995	NA
## Batch14_2024.152	37.5	0.06834928	0.78995	NA
## Batch14_2024.153	40.0	0.06834928	0.79270	NA
## Batch14_2024.154	41.5	0.06834928	0.79270	NA
## Batch14_2024.155	42.8	0.06834928	0.79270	NA
## Batch14_2024.156	43.1	0.06834928	0.79270	NA
## Batch14_2024.157	36.3	0.06834928	0.79070	NA
## Batch14_2024.158	41.3	0.06834928	0.79070	NA
## Batch14_2024.159	42.7	0.06834928	0.79070	NA
## Batch14_2024.160	42.9	0.06834928	0.79070	NA
##	Sample_CH4_Peak	Sample_CH4_ppm	Sample_CO2_Peak	Sample_CO2_ppm
## Batch12_2024.1	59701	10.860764	NA	NA

## Batch12_2024.2	144169	25.988029	NA	NA
## Batch12_2024.3	227122	40.843975	NA	NA
## Batch12_2024.4	310759	55.822417	NA	NA
## Batch12_2024.5	45013	8.230309	NA	NA
## Batch12_2024.6	82035	14.860531	NA	NA
## Batch12_2024.7	114774	20.723716	NA	NA
## Batch12_2024.8	144773	26.096199	NA	NA
## Batch12_2024.9	51940	9.470856	NA	NA
## Batch12_2024.10	84241	15.255601	NA	NA
## Batch12_2024.11	112839	20.377180	NA	NA
## Batch12_2024.12	137742	24.837026	NA	NA
## Batch12_2024.13	70226	12.745673	NA	NA
## Batch12_2024.14	118334	21.361272	NA	NA
## Batch12_2024.15	159713	28.771784	NA	NA
## Batch12_2024.16	207247	37.284587	NA	NA
## Batch12_2024.17	259633	46.666328	NA	NA
## Batch12_2024.18	410334	73.655176	NA	NA
## Batch12_2024.19	573167	102.816727	NA	NA
## Batch12_2024.20	746536	133.865158	NA	NA
## Batch12_2024.21	17655	3.330800	NA	NA
## Batch12_2024.22	50011	9.125394	NA	NA
## Batch12_2024.23	78440	14.216707	NA	NA
## Batch12_2024.24	106280	19.202537	NA	NA
## Batch12_2024.25	16449	3.114819	NA	NA
## Batch12_2024.26	163679	29.482050	NA	NA
## Batch12_2024.27	306079	54.984282	NA	NA
## Batch12_2024.28	442606	79.434726	NA	NA
## Batch12_2024.29	14666	2.795504	NA	NA
## Batch12_2024.30	37138	6.819985	NA	NA
## Batch12_2024.31	53796	9.803245	NA	NA
## Batch12_2024.32	75401	13.672457	NA	NA
## Batch12_2024.33	13250	2.541914	NA	NA
## Batch12_2024.34	64316	11.687258	NA	NA
## Batch12_2024.35	112706	20.353361	NA	NA
## Batch12_2024.36	159532	28.739369	NA	NA
## Batch12_2024.37	15430	2.932328	NA	NA
## Batch12_2024.38	139978	25.237468	NA	NA
## Batch12_2024.39	260619	46.842909	NA	NA
## Batch12_2024.40	386882	69.455187	NA	NA
## Batch12_2024.41	93759	16.960167	NA	NA
## Batch12_2024.42	295171	53.030782	NA	NA
## Batch12_2024.43	482628	86.602215	NA	NA
## Batch12_2024.44	657941	117.998794	NA	NA
## Batch12_2024.45	83602	15.141163	NA	NA
## Batch12_2024.46	175826	31.657440	NA	NA
## Batch12_2024.47	263411	47.342925	NA	NA
## Batch12_2024.48	343467	61.680051	NA	NA
## Batch12_2024.49	39227	7.194102	NA	NA
## Batch12_2024.50	64061	11.641591	NA	NA
## Batch12_2024.51	87789	15.891008	NA	NA
## Batch12_2024.52	110621	19.979961	NA	NA
## Batch12_2024.53	79580	14.420868	NA	NA
## Batch12_2024.54	132422	23.884274	NA	NA
## Batch12_2024.55	193813	34.878709	NA	NA

## Batch12_2024.56	247312	44.459776	NA	NA
## Batch12_2024.57	85507	15.482327	NA	NA
## Batch12_2024.58	134387	24.236183	NA	NA
## Batch12_2024.59	189081	34.031261	NA	NA
## Batch12_2024.60	233531	41.991754	NA	NA
## Batch12_2024.61	19762	3.708140	NA	NA
## Batch12_2024.62	68369	12.413105	NA	NA
## Batch12_2024.63	110988	20.045687	NA	NA
## Batch12_2024.64	151773	27.349820	NA	NA
## Batch12_2024.65	16788	3.175530	NA	NA
## Batch12_2024.66	133189	24.021635	NA	NA
## Batch12_2024.67	241222	43.369126	NA	NA
## Batch12_2024.68	351103	63.047572	NA	NA
## Batch12_2024.69	15137	2.879855	NA	NA
## Batch12_2024.70	58962	10.728417	NA	NA
## Batch12_2024.71	100696	18.202506	NA	NA
## Batch12_2024.72	142020	25.603167	NA	NA
## Batch12_2024.73	14840	2.826665	NA	NA
## Batch12_2024.74	59950	10.905357	NA	NA
## Batch12_2024.75	103659	18.733145	NA	NA
## Batch12_2024.76	145397	26.207950	NA	NA
## Batch12_2024.77	13342	2.558390	NA	NA
## Batch12_2024.78	51088	9.318273	NA	NA
## Batch12_2024.79	86432	15.647984	NA	NA
## Batch12_2024.80	119801	21.623995	NA	NA
## Batch14_2024.81	84743	15.034195	NA	NA
## Batch14_2024.82	236721	41.747879	NA	NA
## Batch14_2024.83	426998	75.193507	NA	NA
## Batch14_2024.84	604575	106.406813	NA	NA
## Batch14_2024.85	64968	11.558277	NA	NA
## Batch14_2024.86	190049	33.544185	NA	NA
## Batch14_2024.87	343349	60.490241	NA	NA
## Batch14_2024.88	498803	87.814913	NA	NA
## Batch14_2024.89	93336	16.544616	NA	NA
## Batch14_2024.90	229867	40.543128	NA	NA
## Batch14_2024.91	382312	67.338898	NA	NA
## Batch14_2024.92	521456	91.796707	NA	NA
## Batch14_2024.93	61272	10.908619	NA	NA
## Batch14_2024.94	208670	36.817261	NA	NA
## Batch14_2024.95	382880	67.438738	NA	NA
## Batch14_2024.96	550141	96.838766	NA	NA
## Batch14_2024.97	62594	11.140991	NA	NA
## Batch14_2024.98	205119	36.193089	NA	NA
## Batch14_2024.99	348819	61.451721	NA	NA
## Batch14_2024.100	475145	83.656467	NA	NA
## Batch14_2024.101	48106	8.594387	NA	NA
## Batch14_2024.102	149608	26.435734	NA	NA
## Batch14_2024.103	270450	47.676539	NA	NA
## Batch14_2024.104	385629	67.921939	NA	NA
## Batch14_2024.105	58467	10.415575	NA	NA
## Batch14_2024.106	174309	30.777512	NA	NA
## Batch14_2024.107	310583	54.730851	NA	NA
## Batch14_2024.108	437127	76.973915	NA	NA
## Batch14_2024.109	66194	11.773775	NA	NA

## Batch14_2024.110	158631	28.021737	NA	NA
## Batch14_2024.111	259664	45.780647	NA	NA
## Batch14_2024.112	352226	62.050581	NA	NA
## Batch14_2024.113	104024	18.423282	NA	NA
## Batch14_2024.114	275683	48.596361	NA	NA
## Batch14_2024.115	458974	80.814036	NA	NA
## Batch14_2024.116	632065	111.238822	NA	NA
## Batch14_2024.117	107580	19.048332	NA	NA
## Batch14_2024.118	309473	54.535743	NA	NA
## Batch14_2024.119	505344	88.964646	NA	NA
## Batch14_2024.120	756101	133.041047	NA	NA
## Batch14_2024.121	48004	8.576458	NA	NA
## Batch14_2024.122	132319	23.396789	NA	NA
## Batch14_2024.123	294552	51.913028	NA	NA
## Batch14_2024.124	389539	68.609212	NA	NA
## Batch14_2024.125	59909	10.669040	NA	NA
## Batch14_2024.126	155927	27.546446	NA	NA
## Batch14_2024.127	315294	55.558919	NA	NA
## Batch14_2024.128	510359	89.846150	NA	NA
## Batch14_2024.129	68567	12.190886	NA	NA
## Batch14_2024.130	144589	25.553528	NA	NA
## Batch14_2024.131	269624	47.531350	NA	NA
## Batch14_2024.132	339633	59.837067	NA	NA
## Batch14_2024.133	90550	16.054911	NA	NA
## Batch14_2024.134	206925	36.510536	NA	NA
## Batch14_2024.135	386925	68.149741	NA	NA
## Batch14_2024.136	538881	94.859557	NA	NA
## Batch14_2024.137	95497	16.924462	NA	NA
## Batch14_2024.138	176183	31.106911	NA	NA
## Batch14_2024.139	342502	60.341361	NA	NA
## Batch14_2024.140	435742	76.730469	NA	NA
## Batch14_2024.141	46697	8.346722	NA	NA
## Batch14_2024.142	137366	24.283917	NA	NA
## Batch14_2024.143	255705	45.084761	NA	NA
## Batch14_2024.144	363809	64.086564	NA	NA
## Batch14_2024.145	44423	7.947014	NA	NA
## Batch14_2024.146	131822	23.309429	NA	NA
## Batch14_2024.147	260479	45.923903	NA	NA
## Batch14_2024.148	359000	63.241270	NA	NA
## Batch14_2024.149	87014	15.433377	NA	NA
## Batch14_2024.150	208288	36.750115	NA	NA
## Batch14_2024.151	315120	55.528335	NA	NA
## Batch14_2024.152	501953	88.368599	NA	NA
## Batch14_2024.153	67372	11.980836	NA	NA
## Batch14_2024.154	191454	33.791146	NA	NA
## Batch14_2024.155	326728	57.568712	NA	NA
## Batch14_2024.156	464688	81.818405	NA	NA
## Batch14_2024.157	79001	14.024905	NA	NA
## Batch14_2024.158	181124	31.975408	NA	NA
## Batch14_2024.159	294139	51.840434	NA	NA
## Batch14_2024.160	396874	69.898510	NA	NA
##	Sample_N20_Peak	Sample_N20_ppm	Sample_Gas1_Peak	
## Batch12_2024.1	40962	0.3164698	NA	
## Batch12_2024.2	40955	0.3163928	NA	

## Batch12_2024.3	40341	0.3096409	NA
## Batch12_2024.4	41718	0.3247831	NA
## Batch12_2024.5	40518	0.3115873	NA
## Batch12_2024.6	39510	0.3005028	NA
## Batch12_2024.7	40220	0.3083103	NA
## Batch12_2024.8	40269	0.3088492	NA
## Batch12_2024.9	41504	0.3224299	NA
## Batch12_2024.10	40712	0.3137206	NA
## Batch12_2024.11	42109	0.3290828	NA
## Batch12_2024.12	40429	0.3106086	NA
## Batch12_2024.13	41026	0.3171735	NA
## Batch12_2024.14	40463	0.3109825	NA
## Batch12_2024.15	40304	0.3092341	NA
## Batch12_2024.16	41072	0.3176794	NA
## Batch12_2024.17	40182	0.3078925	NA
## Batch12_2024.18	41175	0.3188120	NA
## Batch12_2024.19	41496	0.3223419	NA
## Batch12_2024.20	41173	0.3187900	NA
## Batch12_2024.21	41527	0.3226828	NA
## Batch12_2024.22	41165	0.3187021	NA
## Batch12_2024.23	41749	0.3251240	NA
## Batch12_2024.24	41118	0.3181852	NA
## Batch12_2024.25	41487	0.3222429	NA
## Batch12_2024.26	41687	0.3244422	NA
## Batch12_2024.27	40372	0.3099818	NA
## Batch12_2024.28	41408	0.3213742	NA
## Batch12_2024.29	39280	0.2979736	NA
## Batch12_2024.30	42572	0.3341741	NA
## Batch12_2024.31	41743	0.3250580	NA
## Batch12_2024.32	41325	0.3204615	NA
## Batch12_2024.33	42515	0.3335473	NA
## Batch12_2024.34	41613	0.3236285	NA
## Batch12_2024.35	41854	0.3262786	NA
## Batch12_2024.36	41756	0.3252010	NA
## Batch12_2024.37	41572	0.3231776	NA
## Batch12_2024.38	40594	0.3124230	NA
## Batch12_2024.39	40583	0.3123021	NA
## Batch12_2024.40	42418	0.3324807	NA
## Batch12_2024.41	42198	0.3300614	NA
## Batch12_2024.42	42378	0.3320408	NA
## Batch12_2024.43	41735	0.3249701	NA
## Batch12_2024.44	40867	0.3154251	NA
## Batch12_2024.45	40434	0.3106636	NA
## Batch12_2024.46	42090	0.3288738	NA
## Batch12_2024.47	42074	0.3286979	NA
## Batch12_2024.48	40573	0.3121921	NA
## Batch12_2024.49	39929	0.3051104	NA
## Batch12_2024.50	41496	0.3223419	NA
## Batch12_2024.51	41444	0.3217701	NA
## Batch12_2024.52	40685	0.3134237	NA
## Batch12_2024.53	40699	0.3135777	NA
## Batch12_2024.54	39834	0.3040657	NA
## Batch12_2024.55	40010	0.3060011	NA
## Batch12_2024.56	40728	0.3138966	NA

## Batch12_2024.57	40651	0.3130498	NA
## Batch12_2024.58	41069	0.3176464	NA
## Batch12_2024.59	40900	0.3157880	NA
## Batch12_2024.60	40733	0.3139516	NA
## Batch12_2024.61	41891	0.3266855	NA
## Batch12_2024.62	40459	0.3109385	NA
## Batch12_2024.63	40833	0.3150512	NA
## Batch12_2024.64	40002	0.3059131	NA
## Batch12_2024.65	41711	0.3247061	NA
## Batch12_2024.66	41242	0.3195488	NA
## Batch12_2024.67	41370	0.3209563	NA
## Batch12_2024.68	42500	0.3333824	NA
## Batch12_2024.69	41343	0.3206594	NA
## Batch12_2024.70	40461	0.3109605	NA
## Batch12_2024.71	39589	0.3013715	NA
## Batch12_2024.72	40661	0.3131598	NA
## Batch12_2024.73	40029	0.3062100	NA
## Batch12_2024.74	40432	0.3106416	NA
## Batch12_2024.75	40901	0.3157990	NA
## Batch12_2024.76	39303	0.2982265	NA
## Batch12_2024.77	40850	0.3152382	NA
## Batch12_2024.78	41482	0.3221879	NA
## Batch12_2024.79	40181	0.3078815	NA
## Batch12_2024.80	41238	0.3195048	NA
## Batch14_2024.81	36840	0.3088018	NA
## Batch14_2024.82	38041	0.3226310	NA
## Batch14_2024.83	39207	0.3360571	NA
## Batch14_2024.84	38374	0.3264654	NA
## Batch14_2024.85	38120	0.3235407	NA
## Batch14_2024.86	37779	0.3196141	NA
## Batch14_2024.87	38058	0.3228267	NA
## Batch14_2024.88	39641	0.3410545	NA
## Batch14_2024.89	37371	0.3149161	NA
## Batch14_2024.90	37933	0.3213874	NA
## Batch14_2024.91	38151	0.3238976	NA
## Batch14_2024.92	38473	0.3276053	NA
## Batch14_2024.93	38474	0.3276169	NA
## Batch14_2024.94	38026	0.3224583	NA
## Batch14_2024.95	38136	0.3237249	NA
## Batch14_2024.96	38773	0.3310598	NA
## Batch14_2024.97	38683	0.3300234	NA
## Batch14_2024.98	39025	0.3339615	NA
## Batch14_2024.99	38302	0.3256363	NA
## Batch14_2024.100	38061	0.3228613	NA
## Batch14_2024.101	38521	0.3281581	NA
## Batch14_2024.102	38252	0.3250606	NA
## Batch14_2024.103	37997	0.3221243	NA
## Batch14_2024.104	38158	0.3239782	NA
## Batch14_2024.105	38134	0.3237019	NA
## Batch14_2024.106	38979	0.3334318	NA
## Batch14_2024.107	39183	0.3357808	NA
## Batch14_2024.108	38263	0.3251873	NA
## Batch14_2024.109	37970	0.3218134	NA
## Batch14_2024.110	38147	0.3238515	NA

## Batch14_2024.111	38514	0.3280774	NA
## Batch14_2024.112	37741	0.3191766	NA
## Batch14_2024.113	37052	0.3112429	NA
## Batch14_2024.114	37853	0.3204662	NA
## Batch14_2024.115	38023	0.3224237	NA
## Batch14_2024.116	38007	0.3222395	NA
## Batch14_2024.117	36974	0.3103448	NA
## Batch14_2024.118	37877	0.3207426	NA
## Batch14_2024.119	37250	0.3135228	NA
## Batch14_2024.120	37451	0.3158373	NA
## Batch14_2024.121	38154	0.3239322	NA
## Batch14_2024.122	38007	0.3222395	NA
## Batch14_2024.123	38130	0.3236558	NA
## Batch14_2024.124	37368	0.3148816	NA
## Batch14_2024.125	37739	0.3191535	NA
## Batch14_2024.126	38789	0.3312440	NA
## Batch14_2024.127	38716	0.3304034	NA
## Batch14_2024.128	37729	0.3190384	NA
## Batch14_2024.129	37717	0.3189002	NA
## Batch14_2024.130	37559	0.3170809	NA
## Batch14_2024.131	38711	0.3303458	NA
## Batch14_2024.132	39013	0.3338233	NA
## Batch14_2024.133	39329	0.3374619	NA
## Batch14_2024.134	38148	0.3238631	NA
## Batch14_2024.135	37890	0.3208923	NA
## Batch14_2024.136	38807	0.3314513	NA
## Batch14_2024.137	38083	0.3231146	NA
## Batch14_2024.138	37063	0.3113696	NA
## Batch14_2024.139	38043	0.3226540	NA
## Batch14_2024.140	38273	0.3253024	NA
## Batch14_2024.141	37892	0.3209153	NA
## Batch14_2024.142	39157	0.3354814	NA
## Batch14_2024.143	37276	0.3138222	NA
## Batch14_2024.144	39697	0.3416994	NA
## Batch14_2024.145	38842	0.3318543	NA
## Batch14_2024.146	38210	0.3245770	NA
## Batch14_2024.147	37835	0.3202590	NA
## Batch14_2024.148	38351	0.3262005	NA
## Batch14_2024.149	37008	0.3107363	NA
## Batch14_2024.150	38415	0.3269375	NA
## Batch14_2024.151	37838	0.3202935	NA
## Batch14_2024.152	38001	0.3221704	NA
## Batch14_2024.153	38070	0.3229649	NA
## Batch14_2024.154	38759	0.3308986	NA
## Batch14_2024.155	38090	0.3231952	NA
## Batch14_2024.156	35822	0.2970798	NA
## Batch14_2024.157	37746	0.3192341	NA
## Batch14_2024.158	38090	0.3231952	NA
## Batch14_2024.159	37199	0.3129356	NA
## Batch14_2024.160	38621	0.3293095	NA
##	Sample_Gas1_ppm	Sample_Gas2_Peak	Sample_Gas2_ppm
## Batch12_2024.1	NA	NA	NA
## Batch12_2024.2	NA	NA	NA
## Batch12_2024.3	NA	NA	NA

## Batch12_2024.4	NA	NA	NA
## Batch12_2024.5	NA	NA	NA
## Batch12_2024.6	NA	NA	NA
## Batch12_2024.7	NA	NA	NA
## Batch12_2024.8	NA	NA	NA
## Batch12_2024.9	NA	NA	NA
## Batch12_2024.10	NA	NA	NA
## Batch12_2024.11	NA	NA	NA
## Batch12_2024.12	NA	NA	NA
## Batch12_2024.13	NA	NA	NA
## Batch12_2024.14	NA	NA	NA
## Batch12_2024.15	NA	NA	NA
## Batch12_2024.16	NA	NA	NA
## Batch12_2024.17	NA	NA	NA
## Batch12_2024.18	NA	NA	NA
## Batch12_2024.19	NA	NA	NA
## Batch12_2024.20	NA	NA	NA
## Batch12_2024.21	NA	NA	NA
## Batch12_2024.22	NA	NA	NA
## Batch12_2024.23	NA	NA	NA
## Batch12_2024.24	NA	NA	NA
## Batch12_2024.25	NA	NA	NA
## Batch12_2024.26	NA	NA	NA
## Batch12_2024.27	NA	NA	NA
## Batch12_2024.28	NA	NA	NA
## Batch12_2024.29	NA	NA	NA
## Batch12_2024.30	NA	NA	NA
## Batch12_2024.31	NA	NA	NA
## Batch12_2024.32	NA	NA	NA
## Batch12_2024.33	NA	NA	NA
## Batch12_2024.34	NA	NA	NA
## Batch12_2024.35	NA	NA	NA
## Batch12_2024.36	NA	NA	NA
## Batch12_2024.37	NA	NA	NA
## Batch12_2024.38	NA	NA	NA
## Batch12_2024.39	NA	NA	NA
## Batch12_2024.40	NA	NA	NA
## Batch12_2024.41	NA	NA	NA
## Batch12_2024.42	NA	NA	NA
## Batch12_2024.43	NA	NA	NA
## Batch12_2024.44	NA	NA	NA
## Batch12_2024.45	NA	NA	NA
## Batch12_2024.46	NA	NA	NA
## Batch12_2024.47	NA	NA	NA
## Batch12_2024.48	NA	NA	NA
## Batch12_2024.49	NA	NA	NA
## Batch12_2024.50	NA	NA	NA
## Batch12_2024.51	NA	NA	NA
## Batch12_2024.52	NA	NA	NA
## Batch12_2024.53	NA	NA	NA
## Batch12_2024.54	NA	NA	NA
## Batch12_2024.55	NA	NA	NA
## Batch12_2024.56	NA	NA	NA
## Batch12_2024.57	NA	NA	NA

## Batch12_2024.58	NA	NA	NA
## Batch12_2024.59	NA	NA	NA
## Batch12_2024.60	NA	NA	NA
## Batch12_2024.61	NA	NA	NA
## Batch12_2024.62	NA	NA	NA
## Batch12_2024.63	NA	NA	NA
## Batch12_2024.64	NA	NA	NA
## Batch12_2024.65	NA	NA	NA
## Batch12_2024.66	NA	NA	NA
## Batch12_2024.67	NA	NA	NA
## Batch12_2024.68	NA	NA	NA
## Batch12_2024.69	NA	NA	NA
## Batch12_2024.70	NA	NA	NA
## Batch12_2024.71	NA	NA	NA
## Batch12_2024.72	NA	NA	NA
## Batch12_2024.73	NA	NA	NA
## Batch12_2024.74	NA	NA	NA
## Batch12_2024.75	NA	NA	NA
## Batch12_2024.76	NA	NA	NA
## Batch12_2024.77	NA	NA	NA
## Batch12_2024.78	NA	NA	NA
## Batch12_2024.79	NA	NA	NA
## Batch12_2024.80	NA	NA	NA
## Batch14_2024.81	NA	NA	NA
## Batch14_2024.82	NA	NA	NA
## Batch14_2024.83	NA	NA	NA
## Batch14_2024.84	NA	NA	NA
## Batch14_2024.85	NA	NA	NA
## Batch14_2024.86	NA	NA	NA
## Batch14_2024.87	NA	NA	NA
## Batch14_2024.88	NA	NA	NA
## Batch14_2024.89	NA	NA	NA
## Batch14_2024.90	NA	NA	NA
## Batch14_2024.91	NA	NA	NA
## Batch14_2024.92	NA	NA	NA
## Batch14_2024.93	NA	NA	NA
## Batch14_2024.94	NA	NA	NA
## Batch14_2024.95	NA	NA	NA
## Batch14_2024.96	NA	NA	NA
## Batch14_2024.97	NA	NA	NA
## Batch14_2024.98	NA	NA	NA
## Batch14_2024.99	NA	NA	NA
## Batch14_2024.100	NA	NA	NA
## Batch14_2024.101	NA	NA	NA
## Batch14_2024.102	NA	NA	NA
## Batch14_2024.103	NA	NA	NA
## Batch14_2024.104	NA	NA	NA
## Batch14_2024.105	NA	NA	NA
## Batch14_2024.106	NA	NA	NA
## Batch14_2024.107	NA	NA	NA
## Batch14_2024.108	NA	NA	NA
## Batch14_2024.109	NA	NA	NA
## Batch14_2024.110	NA	NA	NA
## Batch14_2024.111	NA	NA	NA

## Batch14_2024.112	NA	NA	NA
## Batch14_2024.113	NA	NA	NA
## Batch14_2024.114	NA	NA	NA
## Batch14_2024.115	NA	NA	NA
## Batch14_2024.116	NA	NA	NA
## Batch14_2024.117	NA	NA	NA
## Batch14_2024.118	NA	NA	NA
## Batch14_2024.119	NA	NA	NA
## Batch14_2024.120	NA	NA	NA
## Batch14_2024.121	NA	NA	NA
## Batch14_2024.122	NA	NA	NA
## Batch14_2024.123	NA	NA	NA
## Batch14_2024.124	NA	NA	NA
## Batch14_2024.125	NA	NA	NA
## Batch14_2024.126	NA	NA	NA
## Batch14_2024.127	NA	NA	NA
## Batch14_2024.128	NA	NA	NA
## Batch14_2024.129	NA	NA	NA
## Batch14_2024.130	NA	NA	NA
## Batch14_2024.131	NA	NA	NA
## Batch14_2024.132	NA	NA	NA
## Batch14_2024.133	NA	NA	NA
## Batch14_2024.134	NA	NA	NA
## Batch14_2024.135	NA	NA	NA
## Batch14_2024.136	NA	NA	NA
## Batch14_2024.137	NA	NA	NA
## Batch14_2024.138	NA	NA	NA
## Batch14_2024.139	NA	NA	NA
## Batch14_2024.140	NA	NA	NA
## Batch14_2024.141	NA	NA	NA
## Batch14_2024.142	NA	NA	NA
## Batch14_2024.143	NA	NA	NA
## Batch14_2024.144	NA	NA	NA
## Batch14_2024.145	NA	NA	NA
## Batch14_2024.146	NA	NA	NA
## Batch14_2024.147	NA	NA	NA
## Batch14_2024.148	NA	NA	NA
## Batch14_2024.149	NA	NA	NA
## Batch14_2024.150	NA	NA	NA
## Batch14_2024.151	NA	NA	NA
## Batch14_2024.152	NA	NA	NA
## Batch14_2024.153	NA	NA	NA
## Batch14_2024.154	NA	NA	NA
## Batch14_2024.155	NA	NA	NA
## Batch14_2024.156	NA	NA	NA
## Batch14_2024.157	NA	NA	NA
## Batch14_2024.158	NA	NA	NA
## Batch14_2024.159	NA	NA	NA
## Batch14_2024.160	NA	NA	NA
##			
	Sample_Gas3_Peak	Sample_Gas3_ppm	
## Batch12_2024.1	NA	NA	
## Batch12_2024.2	NA	NA	
## Batch12_2024.3	NA	NA	
## Batch12_2024.4	NA	NA	

## Batch12_2024.5	NA	NA
## Batch12_2024.6	NA	NA
## Batch12_2024.7	NA	NA
## Batch12_2024.8	NA	NA
## Batch12_2024.9	NA	NA
## Batch12_2024.10	NA	NA
## Batch12_2024.11	NA	NA
## Batch12_2024.12	NA	NA
## Batch12_2024.13	NA	NA
## Batch12_2024.14	NA	NA
## Batch12_2024.15	NA	NA
## Batch12_2024.16	NA	NA
## Batch12_2024.17	NA	NA
## Batch12_2024.18	NA	NA
## Batch12_2024.19	NA	NA
## Batch12_2024.20	NA	NA
## Batch12_2024.21	NA	NA
## Batch12_2024.22	NA	NA
## Batch12_2024.23	NA	NA
## Batch12_2024.24	NA	NA
## Batch12_2024.25	NA	NA
## Batch12_2024.26	NA	NA
## Batch12_2024.27	NA	NA
## Batch12_2024.28	NA	NA
## Batch12_2024.29	NA	NA
## Batch12_2024.30	NA	NA
## Batch12_2024.31	NA	NA
## Batch12_2024.32	NA	NA
## Batch12_2024.33	NA	NA
## Batch12_2024.34	NA	NA
## Batch12_2024.35	NA	NA
## Batch12_2024.36	NA	NA
## Batch12_2024.37	NA	NA
## Batch12_2024.38	NA	NA
## Batch12_2024.39	NA	NA
## Batch12_2024.40	NA	NA
## Batch12_2024.41	NA	NA
## Batch12_2024.42	NA	NA
## Batch12_2024.43	NA	NA
## Batch12_2024.44	NA	NA
## Batch12_2024.45	NA	NA
## Batch12_2024.46	NA	NA
## Batch12_2024.47	NA	NA
## Batch12_2024.48	NA	NA
## Batch12_2024.49	NA	NA
## Batch12_2024.50	NA	NA
## Batch12_2024.51	NA	NA
## Batch12_2024.52	NA	NA
## Batch12_2024.53	NA	NA
## Batch12_2024.54	NA	NA
## Batch12_2024.55	NA	NA
## Batch12_2024.56	NA	NA
## Batch12_2024.57	NA	NA
## Batch12_2024.58	NA	NA

## Batch12_2024.59	NA	NA
## Batch12_2024.60	NA	NA
## Batch12_2024.61	NA	NA
## Batch12_2024.62	NA	NA
## Batch12_2024.63	NA	NA
## Batch12_2024.64	NA	NA
## Batch12_2024.65	NA	NA
## Batch12_2024.66	NA	NA
## Batch12_2024.67	NA	NA
## Batch12_2024.68	NA	NA
## Batch12_2024.69	NA	NA
## Batch12_2024.70	NA	NA
## Batch12_2024.71	NA	NA
## Batch12_2024.72	NA	NA
## Batch12_2024.73	NA	NA
## Batch12_2024.74	NA	NA
## Batch12_2024.75	NA	NA
## Batch12_2024.76	NA	NA
## Batch12_2024.77	NA	NA
## Batch12_2024.78	NA	NA
## Batch12_2024.79	NA	NA
## Batch12_2024.80	NA	NA
## Batch14_2024.81	NA	NA
## Batch14_2024.82	NA	NA
## Batch14_2024.83	NA	NA
## Batch14_2024.84	NA	NA
## Batch14_2024.85	NA	NA
## Batch14_2024.86	NA	NA
## Batch14_2024.87	NA	NA
## Batch14_2024.88	NA	NA
## Batch14_2024.89	NA	NA
## Batch14_2024.90	NA	NA
## Batch14_2024.91	NA	NA
## Batch14_2024.92	NA	NA
## Batch14_2024.93	NA	NA
## Batch14_2024.94	NA	NA
## Batch14_2024.95	NA	NA
## Batch14_2024.96	NA	NA
## Batch14_2024.97	NA	NA
## Batch14_2024.98	NA	NA
## Batch14_2024.99	NA	NA
## Batch14_2024.100	NA	NA
## Batch14_2024.101	NA	NA
## Batch14_2024.102	NA	NA
## Batch14_2024.103	NA	NA
## Batch14_2024.104	NA	NA
## Batch14_2024.105	NA	NA
## Batch14_2024.106	NA	NA
## Batch14_2024.107	NA	NA
## Batch14_2024.108	NA	NA
## Batch14_2024.109	NA	NA
## Batch14_2024.110	NA	NA
## Batch14_2024.111	NA	NA
## Batch14_2024.112	NA	NA

## Batch14_2024.113	NA	NA
## Batch14_2024.114	NA	NA
## Batch14_2024.115	NA	NA
## Batch14_2024.116	NA	NA
## Batch14_2024.117	NA	NA
## Batch14_2024.118	NA	NA
## Batch14_2024.119	NA	NA
## Batch14_2024.120	NA	NA
## Batch14_2024.121	NA	NA
## Batch14_2024.122	NA	NA
## Batch14_2024.123	NA	NA
## Batch14_2024.124	NA	NA
## Batch14_2024.125	NA	NA
## Batch14_2024.126	NA	NA
## Batch14_2024.127	NA	NA
## Batch14_2024.128	NA	NA
## Batch14_2024.129	NA	NA
## Batch14_2024.130	NA	NA
## Batch14_2024.131	NA	NA
## Batch14_2024.132	NA	NA
## Batch14_2024.133	NA	NA
## Batch14_2024.134	NA	NA
## Batch14_2024.135	NA	NA
## Batch14_2024.136	NA	NA
## Batch14_2024.137	NA	NA
## Batch14_2024.138	NA	NA
## Batch14_2024.139	NA	NA
## Batch14_2024.140	NA	NA
## Batch14_2024.141	NA	NA
## Batch14_2024.142	NA	NA
## Batch14_2024.143	NA	NA
## Batch14_2024.144	NA	NA
## Batch14_2024.145	NA	NA
## Batch14_2024.146	NA	NA
## Batch14_2024.147	NA	NA
## Batch14_2024.148	NA	NA
## Batch14_2024.149	NA	NA
## Batch14_2024.150	NA	NA
## Batch14_2024.151	NA	NA
## Batch14_2024.152	NA	NA
## Batch14_2024.153	NA	NA
## Batch14_2024.154	NA	NA
## Batch14_2024.155	NA	NA
## Batch14_2024.156	NA	NA
## Batch14_2024.157	NA	NA
## Batch14_2024.158	NA	NA
## Batch14_2024.159	NA	NA
## Batch14_2024.160	NA	NA

PPM to flux

```
Final_data <- ppm2flux(sample_ppm, CH4_mass = 16, N2O_mass = 44, Timesteps = 4, Diagnostics = FALSE)
```

```
sample_ppm_check <- read.csv("../data/Input_samples_ppm.csv")  
check_data <- ppm2flux(sample_ppm_check, CH4_mass = 16, N2O_mass = 44, Timesteps = 4, Diagnostics = FALSE)  
#generally looks okay!
```